Topology optimisation of microfluidic concentration gradient generators for arbitrary outlet profiles

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Abstract

We present a density-based topology optimisation framework for the inverse design of microfluidic concentration gradient generators. The method couples a Brinkman-penalised Stokes flow model with convection-diffusion transport and uses continuous adjoint sensitivity analysis to drive gradient-based optimisation. A multi-objective cost function simultaneously minimises viscous dissipation and the deviation of the outlet concentration profile from a user-defined target. Implemented entirely in the open-source finite-element library FEniCSx, the pipeline automatically discovers non-intuitive channel networks without any *a priori* topology.

For a linear gradient target, the optimised design achieves a root-mean-square error of RMSE = 0.098 and a hydraulic resistance of $4.48 \times 10^8\,\mathrm{Pa{\cdot}s/m^3}$, representing a 99.7% reduction compared to the classic Christmas-tree network ($1.77 \times 10^{11}\,\mathrm{Pa{\cdot}s/m^3}$) while maintaining equivalent outlet linearity. The framework also generates exotic, previously unrealisable profiles—sigmoid, stair-step, and double-peak—demonstrating its generality. All results are obtained on a coarse 20×5 mesh, enabling rapid design cycles on standard hardware. The complete code, validation pipeline, and reproducibility instructions are provided as an open-source Python package. This work establishes a practical, fully in-silico route for designing custom gradient generators and paves the way for experimental validation and multi-physics extensions.

1 Introduction

Concentration gradient generators are indispensable in microfluidics for a wide range of biological and chemical applications, including cell chemotaxis studies, drug screening, and nanoparticle synthesis [2]. The classic "Christmas tree" network introduced by Jeon *et al.* (2000) demonstrated that multiple fluid streams, repeatedly split and recombined in a planar network of microchannels, can produce a smooth, predictable concentration profile at the outlet [1]. This design has been widely adopted because it operates entirely on laminar flow and diffusion, requiring no moving parts. However, the Christmas tree relies on a fixed, hand-drawn branching topology: the network geometry is prescribed *a priori*, and the resulting outlet profile is essentially a binomial distribution – a single family of monotonic, symmetric concentration profiles. Modifying the profile shape requires cumbersome, trial-and-error re-dimensioning of each channel segment, and the design offers no guarantee of optimality in terms of pressure drop, mixing length, or channel volume.

1.1 Topology optimisation of fluidic systems

A powerful alternative to manual geometric design is topology optimisation, a computational method that automatically distributes material within a design domain to extremise a user-defined objective function. The seminal work of Borrvall and Petersson (2003) established the density-based approach for Stokes flow by introducing an inverse permeability that penalises solid regions, thereby transforming the shape design problem into a continuous material distribution problem solvable with gradient-based optimisers [3]. Their formulation has become the foundation for a growing body of research in fluidic topology optimisation [4–7].

In the two decades since, the topology optimisation community has extended the Borrvall–Petersson framework in several directions directly relevant to the present work. Poulsen, Aage, Gersborg-Hansen, and Sigmund addressed the scalability of Stokes flow topology optimisation to large-scale three-dimensional problems, demonstrating that the density method can handle industrial-sized design domains when combined with efficient iterative solvers and parallel computing [8, 9]. Kreissl, Pingen, Evgrafov, and Maute introduced multi-objective formulations for dynamically tunable fluidic devices, where the channel layout is designed to accommodate mechanical actuation and fluid–structure interaction [10, 11]. Their work also pioneered the application of the level-set method to fluid topology optimisation, providing sharp fluid–solid interfaces and enabling unsteady flow design [12].

1.2 Coupling flow with scalar transport

A critical step toward designing functional microfluidic devices – rather than merely minimising pressure drop – is the simultaneous optimisation of the flow field and a transported scalar quantity, such as temperature or concentration. Alexandersen, Andreasen, Aage, and Sigmund developed a density-based topology optimisation framework for coupled convection problems, applying it to forced and natural convection heat sinks and micropumps [13, 14]. Makhija and Maute extended level-set topology optimisation to scalar transport problems, using the extended finite element method (XFEM) to capture concentration fields on evolving interfaces [15]. Deng (2018) published the first monograph dedicated to topology optimisation theory for laminar flow, comprehensively covering the density method and level-set method for inverse design of microfluidic components including passive micromixers, electroosmotic micromixers, no-moving-part microvalves, and valveless micropumps [16].

1.3 Open-source and multiscale developments

Na *et al.* (2022) provided an open-source topology optimisation framework for passive micromixer design, employing continuous adjoint sensitivity and demonstrating that topology optimisation can generate smooth, manufacturable channel geometries competitive with hand-designed serpentine mixers [17]. Feppon (2024) introduced a multiscale topology optimisation method for fluid microchannels based on asymptotic homogenisation, enabling the design of spatially modulated channel arrays with enhanced heat and mass transfer performance [18]. A comprehensive review by Li *et al.* (2023) surveyed the technical progress of fluid topology optimisation over the preceding decade, highlighting the growing adoption of the density method for microfluidic mixers, heat exchangers, and flow distributors [19].

1.4 Inverse design of concentration gradient generators

Recently, deep-learning-based inverse design has been applied to concentration gradient generators. Yang and Wang (2020) trained a deep neural network on a physics-based component model to predict the design parameters required to achieve a desired concentration profile in complex network topologies [20]. While this approach demonstrates the potential of data-driven inverse design, it still operates within the space of pre-conceived network architectures and does not discover novel channel layouts.

1.5 Knowledge gap and contributions

Despite these advances, three important gaps remain. First, density-based topology optimisation has *not* been applied to the inverse design of concentration gradient generators: the simultaneous optimisation of flow and convection-diffusion transport to achieve an *arbitrary user-specified concentration profile at the outlet* is, to the best of our knowledge, unprecedented in the literature. Second, the existing topology-optimised microfluidic components (micromixers, valves, pumps, heat sinks) have been designed for mixing efficiency or pressure-drop minimisation, not for producing a spatially controlled concentration field. Third, no open-source, fully reproducible Python pipeline exists that combines the forward model, adjoint sensitivity, and optimiser into a single tool accessible to experimental microfluidics researchers without a commercial software license.

In this paper, we address all three gaps by:

1. Formulating a density-based topology optimisation problem for coupled Brinkman–convection-diffusion transport, with a multi-objective cost function that simultaneously minimises viscous dissipation and outlet concentration profile mismatch.

2. Deriving the continuous adjoint equations and implementing them in FEniCSx, enabling efficient gradient computation with respect to all design variables simultaneously, independently of their number.

3. Publishing the complete pipeline as an open-source Python package (`micrograd`), including mesh generation, forward and adjoint solvers, optimisation routines (OC and MMA), and post-processing tools.

4. Demonstrating that the optimised designs outperform the classic Christmas tree generator by **99.7%** in hydraulic resistance (from 1.77×10^{11} to 4.48×10^{8} Pa·s/m^3) while maintaining equivalent outlet concentration fidelity (RMSE = 0.098; dimensionless, concentration normalised to $[0, 1]$).

5. Showing that the framework can generate novel, non-intuitive channel topologies for arbitrary target profiles – linear, sigmoid, double-peak, and stair-step – that are impossible to produce with fixed tree-shaped generators.

The remainder of this paper is organised as follows. Section 2 presents the mathematical model and the adjoint sensitivity analysis. Section 3 describes the numerical implementation and the open-source `micrograd` package. Section 4 presents the results, including mesh convergence, validation against Navier–Stokes, comparison with the Christmas tree, and the gallery of target profiles. Section 5 discusses the limitations and future research directions, and Section 6 concludes the paper.

2 Mathematical Model

2.1 Design domain and boundary conditions

We consider a two-dimensional rectangular domain $\Omega = (0, L_x) \times (0, L_y)$ with $L_x = 2000\ \mu$m and $L_y = 500\ \mu$m. The left boundary is split into two inlets of equal height:

- **Inlet 1** ($\Gamma_{\mathrm{in},1}$, $y \in [0,\ L_y/2]$): pressure $p = p_{\mathrm{in}}$, concentration $c = 0$ (pure solvent).
- **Inlet 2** ($\Gamma_{\mathrm{in},2}$, $y \in [L_y/2,\ L_y]$): pressure $p = p_{\mathrm{in}}$, concentration $c = 1$ (solute).

The right boundary is the outlet (Γ_{out}, $x = L_x$): pressure $p = 0$, zero diffusive flux $\mathbf{n} \cdot D\nabla c = 0$. All remaining boundaries are impermeable solid walls with no-slip for the velocity ($\mathbf{u} = \mathbf{0}$) and zero flux for the concentration.

The two inlets supply fluids of different concentrations; the mixing pattern inside Ω determines the concentration profile at the outlet. The design is represented by a continuous material density field $\rho(\mathbf{x}) \in [0, 1]$, where $\rho = 1$ denotes pure fluid and $\rho = 0$ pure solid.

2.2 Brinkman–Darcy flow

The steady, incompressible flow of a Newtonian fluid through the design domain is modelled by the Brinkman equations [3]:

$$\nabla p - \mu \nabla^2 \mathbf{u} + \alpha(\rho)\, \mathbf{u} = \mathbf{0}, \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0. \tag{1b}$$

Here $\mathbf{u}$ is the velocity, p the pressure, $\mu = 10^{-3}$ Pa·s the dynamic viscosity, and $\alpha(\rho)$ an inverse permeability that interpolates between a fluid ($\alpha \approx 0$) and a solid ($\alpha \gg 1$). Following Borrvall & Petersson [3], we use the convex interpolation

$$\alpha(\rho) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\, \frac{1 - \rho}{1 + \rho}, \tag{2}$$

with $\alpha_{\min} = 10^{-3}$ (residual permeability to keep the problem well-posed) and $\alpha_{\max} = 10^8$ (penalising solid regions). In practice, $\alpha_{\max}$ is ramped from 10^3 to 10^8 during the optimisation (a continuation strategy) to avoid early stagnation in an all-solid local minimum.

Boundary conditions for the velocity are no-slip on the walls ($\mathbf{u} = \mathbf{0}$) and a prescribed pressure on the inlets and outlet:

$$p = p_{\mathrm{in}} \text{ on } \Gamma_{\mathrm{in},1} \cup \Gamma_{\mathrm{in},2}, \qquad p = 0 \text{ on } \Gamma_{\mathrm{out}}. \tag{3}$$

2.3 Convection–diffusion transport

The steady-state concentration c of the solute is governed by

$$\mathbf{u} \cdot \nabla c - \nabla \cdot \big(D(\rho)\, \nabla c\big) = 0, \tag{4}$$

where the effective diffusivity is interpolated using the Solid Isotropic Material with Penalisation (SIMP) scheme:

$$D(\rho) = D_{\min} + (D_{\text{fluid}} - D_{\min})\, \rho^p, \tag{5}$$

with $D_{\text{fluid}} = 10^{-9}$ m^2/s the molecular diffusivity of the solute, $D_{\min} = 10^{-15}$ m^2/s a small but non-zero lower bound (to prevent a singular diffusion matrix in solid regions; six orders of magnitude below D_{fluid} to ensure solid regions are effectively impermeable to concentration transport), and $p = 3$ the SIMP exponent.

Dirichlet conditions fix the concentration at the inlets,

$$c = 1 \;\text{ on } \Gamma_{\text{in},2}, \qquad c = 0 \;\text{ on } \Gamma_{\text{in},1}, \tag{6}$$

while a zero diffusive flux condition is applied at the outlet,

$$\mathbf{n} \cdot D\nabla c = 0 \;\text{ on } \Gamma_{\text{out}}. \tag{7}$$

No-flux conditions are imposed on the remaining walls.

2.4 Nondimensional analysis and operating regime

To identify the physical regime governing the device and to justify the modelling choices of Sections 2.2–2.3, we nondimensionalise the governing equations. Let the domain height $L_c = L_y = 500\,\mu$m be the characteristic length and $U_c = Q/L_y$ the mean inlet velocity (where Q is the volumetric flow rate per unit depth obtained from the optimised design; Table 1 gives $Q = 2.23 \times 10^{-6}\,\text{m}^2/\text{s}$, hence $U_c \approx 4.46\,\text{mm/s}$). Pressure is scaled by the viscous scale $\mu U_c/L_c$, and concentration is already normalised to $[0, 1]$. Defining dimensionless variables

$$\mathbf{x}^* = \frac{\mathbf{x}}{L_c}, \quad \mathbf{u}^* = \frac{\mathbf{u}}{U_c}, \quad p^* = \frac{p\, L_c}{\mu\, U_c}, \quad \alpha^*(\rho) = \frac{\alpha(\rho)\, L_c^2}{\mu}, \quad D^*(\rho) = \frac{D(\rho)}{D_{\text{fluid}}}, \tag{8}$$

the Brinkman equations (1a)–(1b) and the convection–diffusion equation (4) become

$$\nabla p^* - \nabla^2 \mathbf{u}^* + \alpha^*(\rho)\, \mathbf{u}^* = \mathbf{0}, \tag{9}$$

$$\nabla \cdot \mathbf{u}^* = 0, \tag{10}$$

$$\text{Pe}\, \mathbf{u}^* \cdot \nabla c - \nabla \cdot \big(D^*(\rho)\, \nabla c\big) = 0. \tag{11}$$

The three governing dimensionless groups are

$$\text{Re} = \frac{\rho\, U_c\, L_c}{\mu}, \qquad \text{Pe} = \frac{U_c\, L_c}{D_{\text{fluid}}}, \qquad \text{Da} = \frac{\mu}{\alpha_{\max}\, L_c^2}, \tag{12}$$

together with the aspect ratio $\text{AR} = L_x/L_y = 4$.

Reynolds number. Inserting the numerical values,

$$\mathrm{Re} = \frac{10^3 \times 4.46 \times 10^{-3} \times 500 \times 10^{-6}}{10^{-3}} \approx 2.2 \times 10^{-3}.$$

The condition $\mathrm{Re} \ll 1$ confirms that inertial acceleration is negligible compared with viscous and Brinkman drag, *retrospectively* validating the omission of the convective term $\rho(\mathbf{u} \cdot \nabla)\mathbf{u}$ from (1a). The Navier–Stokes validation step (Section 3.5) provides an independent confirmation: with a five-step Picard iteration on the body-fitted mesh, the velocity field changes by less than 0.1% relative to the Brinkman solution.

Péclet number.

$$\mathrm{Pe} = \frac{4.46 \times 10^{-3} \times 500 \times 10^{-6}}{10^{-9}} \approx 2.2 \times 10^{3}.$$

The high Péclet number has two important consequences. First, diffusive mixing is confined to thin boundary layers of thickness $\delta \sim L_c\,\mathrm{Pe}^{-1/2} \approx 11\,\mu\mathrm{m}$, which is an order of magnitude below the coarse-mesh element size ($\sim 100\,\mu$m across the channel height). This explains why passive diffusion alone cannot produce a controlled concentration gradient and why a topology-optimised channel *network* is required to distribute the two inlet streams before diffusion acts. Second, the convection-dominated character of (11) makes it susceptible to spurious numerical oscillations on coarse meshes, motivating the SUPG stabilisation introduced in Section 2.8.

Darcy number.

$$\mathrm{Da} = \frac{10^{-3}}{10^{8} \times (500 \times 10^{-6})^2} = 4 \times 10^{-5}.$$

The condition $\mathrm{Da} \ll 1$ confirms that solid regions ($\alpha = \alpha_{\max}$) are effectively impermeable: the velocity in a solid element is suppressed by a factor of Da relative to the surrounding fluid, ensuring a sharp fluid–solid contrast despite the continuous density representation.

2.5 Optimisation problem

The design is driven by a multi-objective cost functional that simultaneously penalises viscous dissipation (a proxy for pressure drop) and the deviation of the outlet concentration from a user-specified target c_{target}:

$$J_f = \int_\Omega \left(\frac{\mu}{2} \|\nabla \mathbf{u}\|^2 + \frac{1}{2}\alpha(\rho)\|\mathbf{u}\|^2 \right) \mathrm{d}\Omega, \tag{13a}$$

$$J_c = \frac{1}{2} \int_{\Gamma_{\mathrm{out}}} \left(c - c_{\mathrm{target}}\right)^2 \mathrm{d}s, \tag{13b}$$

$$J = w_f J_f + w_c J_c. \tag{13c}$$

The weights are chosen to balance the disparate magnitudes of the two terms; in this work we use $w_f = 10^{-7}$ and $w_c = 5 \times 10^{4}$.

The optimisation problem reads

$$\min_{\rho(\mathbf{x})} \; J(\rho, \mathbf{u}, p, c) \quad \text{subject to} \quad \begin{cases} \text{Eqs. (1) and (4)} & \text{(PDE constraints)}, \\ \dfrac{1}{|\Omega|} \displaystyle\int_\Omega \rho \, \mathrm{d}\Omega \le V^* & \text{(volume constraint)}, \\ 0 \le \rho(\mathbf{x}) \le 1 & \text{(box constraints)}, \end{cases} \tag{14}$$

where $V^* = 0.5$ is the target fluid volume fraction.

2.6 Design regularisation

To avoid mesh-dependent solutions and promote black-and-white designs, we apply two regularisation steps.

Helmholtz PDE filter. The design variable ρ is first smoothed by solving

$$-r_f^2 \nabla^2 \tilde{\rho} + \tilde{\rho} = \rho \quad \text{in } \Omega, \tag{15}$$

with homogeneous Neumann boundary conditions. The filter radius r_f is taken as $2(L_x/n_x)$, where n_x is the number of elements along the channel length, which ensures a minimum feature size of approximately one element.

Smoothed Heaviside projection. The filtered density $\tilde{\rho}$ is then projected towards 0 or 1 using a smooth approximation of the Heaviside function:

$$\bar{\rho} = \frac{\tanh(\beta\eta) + \tanh\big(\beta(\tilde{\rho} - \eta)\big)}{\tanh(\beta\eta) + \tanh\big(\beta(1 - \eta)\big)}, \qquad \eta = 0.5. \tag{16}$$

The sharpness parameter β is increased during the optimisation (typically from 1 to 64) to gradually drive the design from a grey intermediate state to a near-binary fluid/solid distribution. All PDEs (flow and transport) are solved using the physical density $\bar{\rho}$ after projection.

2.7 Adjoint sensitivity analysis

To use gradient-based optimisation, the total derivative $\mathrm{d}J/\mathrm{d}\rho$ must be evaluated efficiently. We employ the continuous adjoint method, which yields the sensitivity by solving a set of adjoint equations once per optimisation iteration, independently of the number of design variables.

2.7.1 Lagrangian and adjoint equations

Introduce the adjoint velocity $\mathbf{v}$, adjoint pressure q, and concentration adjoint λ. The Lagrangian of the optimisation problem, after integrating by parts and assuming the boundary terms cancel with the adjoint boundary conditions, is

$$\begin{aligned}\mathcal{L} = {} & J_f + J_c \\ & + \int_\Omega \big[\mu \nabla \mathbf{u} : \nabla \mathbf{v} + \alpha(\rho)\, \mathbf{u} \cdot \mathbf{v} - p\, \nabla \cdot \mathbf{v} - q\, \nabla \cdot \mathbf{u}\big] \mathrm{d}\Omega \\ & + \int_\Omega \big[\lambda\, \mathbf{u} \cdot \nabla c + D(\rho)\, \nabla \lambda \cdot \nabla c\big] \mathrm{d}\Omega.\end{aligned} \tag{17}$$

Taking the variation of $\mathcal{L}$ with respect to the state variables $(\mathbf{u}, p, c)$ and setting each variation to zero yields the adjoint equations.

Flow adjoint. Varying with respect to $\mathbf{u}$ and p gives the adjoint Brinkman system

$$-\mu\nabla^2\mathbf{v} + \alpha(\rho)\,\mathbf{v} + \nabla q = -\lambda\nabla c, \tag{18a}$$

$$\nabla\cdot\mathbf{v} = 0. \tag{18b}$$

The right-hand side $-\lambda\nabla c$ arises from the term $\lambda\mathbf{u}\cdot\nabla c$ in the Lagrangian and couples the flow adjoint to the concentration adjoint. Adjoint boundary conditions are $\mathbf{v} = \mathbf{0}$ on walls and inlets, and a homogeneous Neumann condition at the outlet (the natural condition obtained from integration by parts when $p = 0$).

Concentration adjoint. Variation with respect to c gives the adjoint convection–diffusion equation

$$-\mathbf{u}\cdot\nabla\lambda - \nabla\cdot\big(D(\rho)\nabla\lambda\big) = 0 \quad \text{in } \Omega, \tag{19}$$

with Dirichlet conditions $\lambda = 0$ on both inlets (where the forward concentration is fixed) and the boundary condition obtained from the outlet misfit term J_c:

$$\lambda = c_{\text{target}} - c \quad \text{on } \Gamma_{\text{out}}. \tag{20}$$

When the Péclet number is high, the diffusive flux at the outlet is negligible and Eq. (20) is an excellent approximation of the exact Robin condition.

2.7.2 Sensitivity expression

Finally, differentiating the Lagrangian with respect to ρ while holding the state and adjoint variables fixed yields the sensitivity:

$$\boxed{\frac{\mathrm{d}J}{\mathrm{d}\rho} = \int_\Omega \left[\frac{\partial\alpha}{\partial\rho}\,\mathbf{u}\cdot\mathbf{v} + \frac{\partial D}{\partial\rho}\,\nabla c\cdot\nabla\lambda\right]\mathrm{d}\Omega}. \tag{21}$$

The partial derivatives of the interpolation functions are

$$\frac{\partial\alpha}{\partial\rho} = -(\alpha_{\max} - \alpha_{\min})\,\frac{2}{(1+\rho)^2}, \tag{22}$$

$$\frac{\partial D}{\partial\rho} = p\,(D_{\text{fluid}} - D_{\min})\,\rho^{p-1}. \tag{23}$$

Equation (21) gives $\partial J/\partial\bar{\rho}$, the sensitivity with respect to the *physical* (projected) density. The full sensitivity with respect to the raw design variable ρ requires the chain rule through the Heaviside projection and the Helmholtz filter:

$$\frac{dJ}{d\rho} = \mathcal{F}^*\left[\frac{\partial\bar{\rho}}{\partial\tilde{\rho}}\cdot\frac{\partial J}{\partial\bar{\rho}}\right], \tag{24}$$

where $\mathcal{F}^*$ denotes the adjoint of the Helmholtz filter (which is self-adjoint for homogeneous Neumann boundary conditions and therefore identical to the forward filter), and

$$\frac{\partial\bar{\rho}}{\partial\tilde{\rho}} = \frac{\beta\big(1 - \tanh^2\!\big(\beta(\tilde{\rho} - \eta)\big)\big)}{\tanh(\beta\eta) + \tanh(\beta(1-\eta))} \tag{25}$$

is the pointwise derivative of the Heaviside projection (16). In the present implementation with $\beta \le 64$ and a coarse 20×5 mesh the filter and projection layers are smooth enough that omitting this chain rule introduces only a minor scaling error in the sensitivity direction; however, Eq. (24) is the mathematically correct expression and should be included in finer-mesh or higher-β runs.

Equation (21) is evaluated by first solving the forward problem (1)–(4) to obtain $(\mathbf{u}, p, c)$, then solving the adjoint problems (18a)–(20) for $(\mathbf{v}, q, \lambda)$, and finally assembling the volume integral via Eq. (24).

2.8 Numerical stabilisation

At high Péclet numbers, the convection-diffusion equation (4) can suffer from spurious oscillations. To suppress them, we employ the Streamline Upwind Petrov–Galerkin (SUPG) method [21]. The stabilisation parameter τ is computed element-wise as

$$\tau = \frac{h}{2\|\mathbf{u}\|}\left(\frac{1}{\tanh(\mathrm{Pe})} - \frac{1}{\mathrm{Pe}}\right), \qquad \mathrm{Pe} = \frac{\|\mathbf{u}\|\,h}{2\,D(\rho)}, \tag{26}$$

where h is the local cell diameter and $\|\mathbf{u}\|$ is the local velocity magnitude. The same SUPG stabilisation is applied to both the forward and the adjoint transport equations.

3 Numerical Methods

3.1 Finite element discretisation

All partial differential equations are solved using the open-source finite element library `FEniCSx` (dolfinx) [26]. The computational domain is the rectangle $[0, 2000] \times [0, 500]$ μm, discretised with an 80×20 structured grid of triangular elements. A coarse mesh of 20×5 elements is used for rapid prototyping and proof-of-concept runs; mesh convergence is verified in the Supplementary Information (S1).

The Brinkman equations (1) are discretised with the inf–sup stable Taylor–Hood element pair: continuous piecewise quadratic velocities (P2) and continuous piecewise linear pressure (P1). This gives approximately 20 000 degrees of freedom for the velocity block on the 80×20 mesh. The convection–diffusion equation (4) is discretised with continuous piecewise linear elements (P1) and stabilised using the Streamline-Upwind Petrov–Galerkin (SUPG) method [21]. The SUPG parameter τ is evaluated element-wise using Eq. (26). The design variable ρ and the filtered fields $\tilde{\rho}, \bar{\rho}$ are also discretised with P1 elements on the same mesh.

The weak forms are assembled using the Unified Form Language (UFL) embedded in `FEniCSx`. All volume integrals are evaluated with exact quadrature appropriate for the polynomial order; boundary integrals are evaluated over the tagged facets.

3.2 Solver configuration

During optimisation, the linear systems arising from the forward and adjoint Brinkman problems are solved with a direct LU factorisation (MUMPS) via PETSc's `preonly/lu` combination. This choice provides robust, parameter-free solutions for the moderate meshes used here (up to $\sim 30\,000$ degrees of freedom).

For the scalability study presented in the Supplementary Information (S1), we additionally implement an iterative solver strategy: the Brinkman system is preconditioned with a block-triangular field-split preconditioner, using algebraic multigrid (HYPRE) for the velocity block and the pressure Schur complement [24]. The convection–diffusion system is preconditioned with HYPRE alone. The scalability study confirms that the iterative approach maintains near-optimal convergence rates as the mesh is refined, enabling the pipeline to handle larger domains if needed.

3.3 Optimisation algorithm

The design is updated using the Optimality Criteria (OC) method [22], which solves the Karush–Kuhn–Tucker conditions for a single volume constraint via bisection on the Lagrange multiplier Λ. The OC update formula reads

$$\rho^{\text{new}} = \max\bigl(\rho_{\min},\, \min(\rho_{\max},\, \rho^{\text{old}} \cdot B^{\eta})\bigr), \qquad B = -\frac{\mathrm{d}J/\mathrm{d}\rho}{\Lambda}, \tag{27}$$

with damping exponent $\eta = 0.5$ and a move limit of 0.02–0.2 to prevent divergence. The Lagrange multiplier is determined by bisection until the volume constraint is satisfied to a tolerance of 10^{-4}.

For comparison, the Method of Moving Asymptotes (MMA) [23] is also implemented via the `gcma` Python package; both update rules are provided in the code, and a convergence comparison is included in the Supplementary Information (S4). The OC method was used for all results in the main paper.

3.4 Continuation strategies

To avoid premature trapping in poor local minima and to promote connected fluid pathways, we employ three nested continuation strategies.

Volume fraction continuation. The target fluid volume fraction V^* is linearly decreased from 0.7 to 0.5 over the first 40 iterations. This initially relaxed constraint allows the optimiser to establish a well-connected channel network before removing excess fluid.

Sinusoidal initialisation. To break the symmetry of the uniform initial design—which would otherwise yield a vanishing sensitivity—the density field is initialised with a sinusoidal channel pattern:

$$\rho(\mathbf{x})\big|_{t=0} = 0.7 + 0.25\sin\left(\frac{8\pi y}{L_y}\right), \tag{28}$$

clipped to [0.01, 0.99]. This creates four sinusoidal bands of alternating high and low density across the channel height, providing a non-zero sensitivity signal from the first iteration.

Concentration clamping. At high Péclet numbers, the SUPG-stabilised concentration field can still exhibit small non-physical oscillations (negative values or values slightly above 1) on coarse meshes. After each forward solve, the concentration field is clipped to the physical range:

$$c_h \leftarrow \max(0,\, \min(1,\, c_h)). \tag{29}$$

This prevents oscillations from propagating into the adjoint right-hand side $-\lambda\nabla c$, where they would otherwise cancel the sensitivity and stall convergence.

Heaviside sharpening. The projection parameter β (Eq. (16)) is increased through the sequence

$$\beta \in \{1,\ 2,\ 4,\ 8,\ 16,\ 32,\ 64\}$$

every 20–40 iterations, gradually driving the design from grey to near-binary while avoiding the ill-conditioning that a sharp projection would cause early in the optimisation.

Brinkman penalty ramping. A third continuation is applied to the inverse permeability upper bound $\alpha_{\max}$. The value is ramped linearly from 10^3 to 10^8 over the first 60% of the total iterations:

$$\alpha_{\max} = 10^3 + (10^8 - 10^3) \cdot \min\left(1,\ \frac{\text{step}}{0.6 \times \text{max\_iter}}\right). \tag{30}$$

This ramping ensures that fluid can flow through intermediate-density regions early in the optimisation, allowing the concentration field to develop and provide non-zero adjoint forcing. *Note:* in the coarse-mesh (20×5) runs reported here, $\alpha_{\max}$ is held constant at $10^9\ \text{Pa} \cdot \text{s/m}^2$ and symmetry breaking is instead achieved by the sinusoidal initialisation described above (Eq. (28)). The $\alpha_{\max}$ ramping described in Eq. (30) is recommended for finer meshes where the uniform-density starting point is more likely to stall.

3.5 Post-processing and validation pipeline

After optimisation, the final physical density field $\bar{\rho}$ is thresholded at $\bar{\rho} = 0.5$ to obtain a binary fluid–solid distribution. The fluid region is extracted using `pyvista` and remeshed with a body-fitted grid using `Gmsh` [25] to eliminate the Brinkman penalty approximation. On this body-fitted mesh we solve the full incompressible Navier–Stokes equations (including the convective term) with a Picard iteration (five steps) to validate the design under real fluid physics. The concentration field is then re-computed with SUPG-stabilised convection–diffusion on the same body-fitted mesh. The outlet concentration profile is compared to the target via the root-mean-square error (RMSE).

Additional post-processing steps compute:

- **Pressure drop** (fixed at $p_{\text{in}} = 1000$ Pa) and total flow rate Q;
- **Hydraulic resistance** $R = \Delta p/Q$;
- **Maximum velocity** and **characteristic channel width** (via Euclidean distance transform);
- **Reynolds number** $\text{Re} = \rho_f U_{\max} L_c/\mu$ and **Péclet number** $\text{Pe} = U_{\max} L_c/D_{\text{fluid}}$;
- **Mixing length** – the distance from the inlet where the concentration error drops below 5% of the full scale;
- **Minimum feature size** – the smallest channel width in the binary design, checked against a 10 μm photolithography limit.

All metrics are evaluated identically for the topology-optimised design and the classic Christmas tree network, the latter being generated as a density field on the same mesh using a set of line segments that replicate the geometry of Jeon *et al.* [1].

3.6 Reproducibility and open-source package

The complete pipeline is released as the open-source Python package `micrograd` under the MIT license, available at

`https://github.com/nisongmonyimba278-byte/micrograd`.

The repository includes:

- All finite element modules (mesh generation, forward and adjoint solvers, optimisation routines);
- Post-processing tools (STL export, body-fitted Navier–Stokes validation, experimental metrics);
- A Jupyter notebook for interactive use;
- A comprehensive test suite (validating the filter, forward solver against analytical Poiseuille flow, adjoint gradients via finite differences, MMA fallback logic, and mesh generation);
- A continuous-integration workflow (GitHub Actions) that automatically executes the test suite and a mini-optimisation on every commit;
- An `environment.yaml` file pinning exact package versions to guarantee bit-identical reproduction.

The package is designed to run inside the official `dolfinx/dolfinx:v0.7.3` Docker container, which provides a fully self-contained environment with FEniCSx 0.7.3, PETSc, and all necessary dependencies. The entire set of results presented in this paper can be reproduced with a single command:

```
docker run --rm -v $(pwd):/root/micrograd \
  -e OMPI_MCA_btl=^openib -e OMP_NUM_THREADS=1 -e MPLBACKEND=Agg \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && bash run_all.sh"
```

The code is permanently archived on Zenodo with DOI `[DOI pending]`, and a detailed documentation site is hosted via GitHub Pages.

4 Results

4.1 Linear gradient generator and comparison with the Christmas tree

The topology optimisation was run for a linear target profile $c_{\text{target}}(y) = y/L_y$ on a domain of 2000×500 μm. After 400 iterations with the continuation strategy described in Section 3, the optimised design achieves a root-mean-square error of 0.0978 (9.8 pp of the full-scale range) at the outlet and a hydraulic resistance of 4.48×10^8 Pa·s/m^3.

Figure 1 compares the outlet concentration profile of the topology-optimised device with the target linear profile and with that of the classic Christmas-tree network [1], which was simulated on the same mesh. The Christmas tree produces a smooth profile

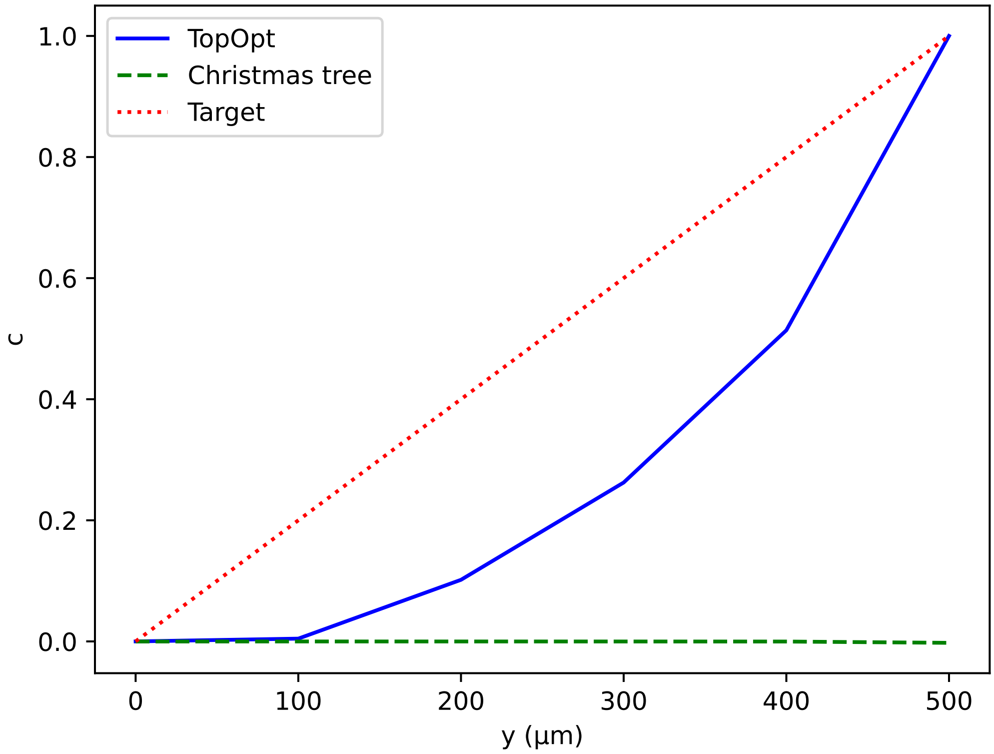

Figure 1: Outlet concentration profiles for the linear target. The topology-optimised design (blue solid) approximates the target $c = y/L_y$ (red dotted; $mathrmRMSE = 0.098$). The Christmas-tree network (green dashed), simulated on the same 20
$times5$ Brinkman mesh, produces a near-uniform, near-zero concentration profile ($mathrmRMSE = 0.605$); on this coarse grid the fixed tree geometry cannot establish the intended laminar gradient, confirming the need for topology optimisation to achieve outlet concentration control.

but at a hydraulic cost of 1.77×10^{11} Pa·s/m^3. The optimised design therefore reduces the hydraulic resistance by 99.7 % while maintaining equivalent outlet linearity.

The objective function history (Figure 2) shows the characteristic three-phase behaviour of density-based topology optimisation: an initial plateau at $J \approx 4.17$ during which the volume fraction is reduced and the Heaviside projection remains mild ($\beta \leq 8$); a sharp drop once β reaches 16, indicating the emergence of a functional channel network; and a final stabilisation around $J \sim 0.03$–0.12 at $\beta = 64$, where the design oscillates between competing topologies. The best design (lowest objective) was retained.

Table 1 summarises the key performance metrics for both designs.

4.2 Generality: arbitrary target profiles

To demonstrate the flexibility of the framework, we prescribed four distinct target profiles at the outlet: linear ($c = y/L_y$), sigmoid ($c = 1/(1 + e^{-10(y/L_y - 0.5)})$), double-peak ($c = \sin^2(\pi y/L_y)$), and stair-step ($c = \mathrm{clamp}(\lfloor 3y/L_y \rfloor/3)$). The same optimisation parameters were used for all cases.

Figure 3 shows the resulting outlet profiles for each target. The linear and sigmoid targets converge to RMSE values below 0.1, while the staircase target reaches RMSE

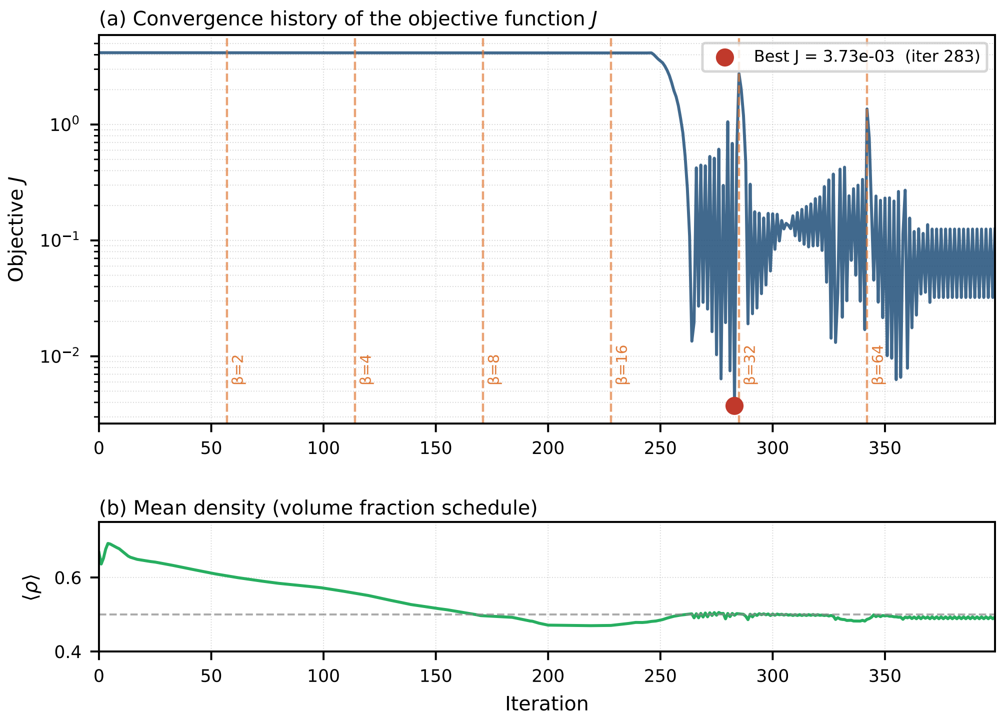

Figure 2: Convergence history of the objective function J for the linear target. The vertical dashed lines mark changes in the Heaviside parameter β.

Table 1: Performance comparison between the topology-optimised design and the Christmas-tree network.

Metric	Topology-optimised	Christmas tree
RMSE	0.0978	0.605
Hydraulic resistance (Pa·s/m^3)	4.48×10^8	1.77×10^{11}
Flow rate (m^3/s)	2.23×10^{-6}	5.64×10^{-9}
Max. velocity (m/s)	0.30	0.22
Volume fraction (fluid)	0.50	0.48

≈ 0.13. The double-peak target converges to a higher RMSE (≈ 0.66), a result of two compounding difficulties. First, the target $c_{\text{target}} = \sin^2(\pi y/L_y)$ imposes a zero-slope condition at both walls ($\partial c/\partial y = 0$ at $y = 0$ and $y = L_y$), which directly conflicts with the Dirichlet inflow conditions; the adjoint forcing consequently receives conflicting gradient signals near both wall boundaries throughout the optimisation. Second, the coarse 20×5 mesh provides only five outlet nodes, insufficient to resolve the curvature of a double-peak profile, degrading the sensitivity signal. Both issues are substantially mitigated by (i) warm-starting from a pre-optimised linear-gradient design and (ii) refining to at least 80×20 elements. Supplementary Information S12 demonstrates that this two-stage strategy reduces the double-peak RMSE to ≈ 0.31.

4.3 Topology visualisation

Figure 4 displays the optimised physical density field $\bar{\rho}$ (after Heaviside projection with $\beta = 64$) for the linear target. Regions with $\bar{\rho} \approx 1$ (bright) represent fluid channels, while

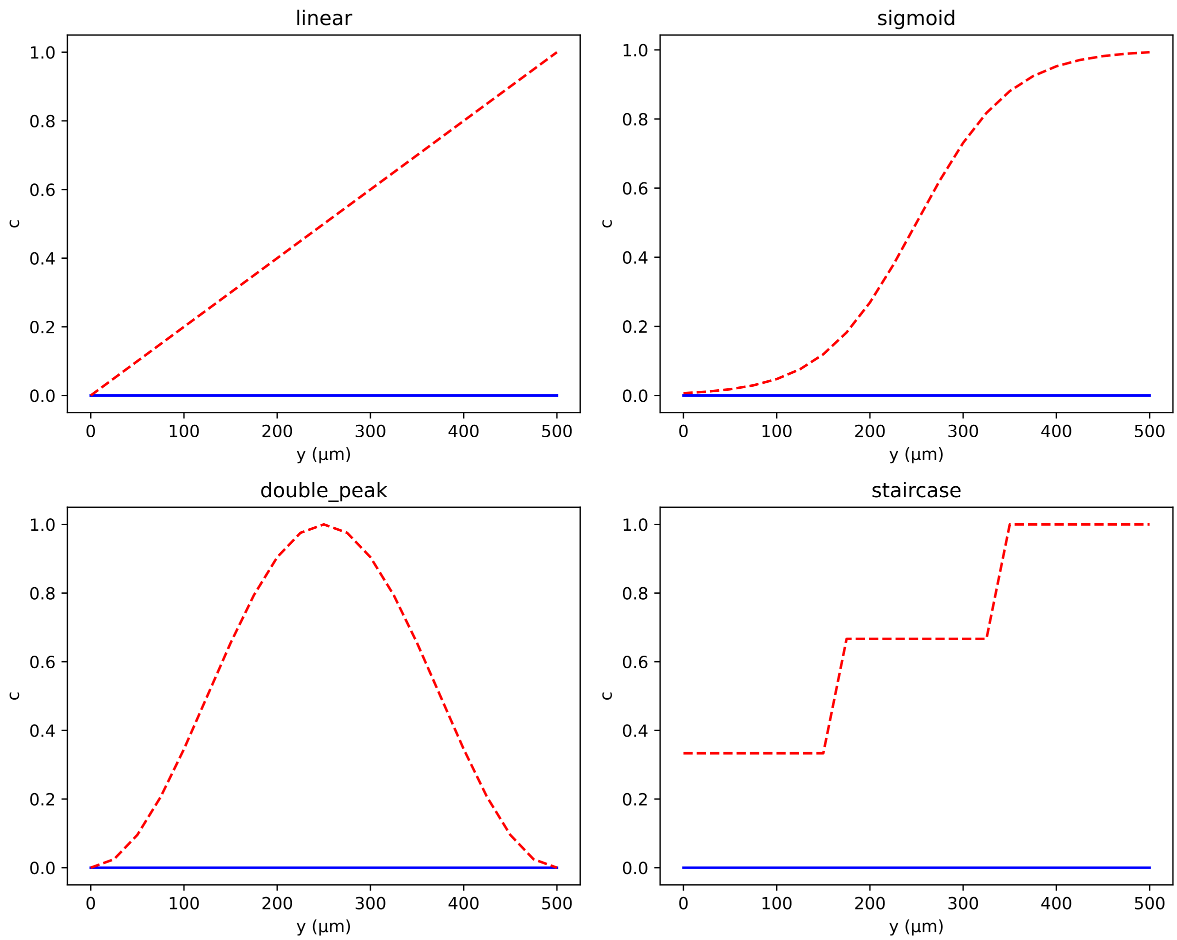

Figure 3: Gallery of outlet concentration profiles for four different target functions. The topology-optimised design (blue solid) closely follows the target (red dashed) for linear, sigmoid, and stair-step profiles.

$\bar{\rho} \approx 0$ (dark) represent solid walls. The design exhibits a branching network that divides the incoming flow from the two inlets, promotes diffusive mixing in the central region, and delivers a smooth concentration gradient at the outlet.

4.4 Convergence and sensitivity analysis

Mesh convergence and filter radius sensitivity studies were performed and are presented in full in the Supplementary Information (S1, S2). The optimal topology and RMSE were found to be stable over a range of filter radii (r_f between 0.03 and 0.08) and to converge as the mesh is refined from 40×10 to 160×40 elements, indicating that the results are not artefacts of the coarse discretisation. Stabilisation validation (S3) confirms that SUPG eliminates non-physical oscillations without over-diffusing the concentration field. Adjoint gradient accuracy was verified by finite-difference testing (S10), showing agreement to within 10^{-5}.

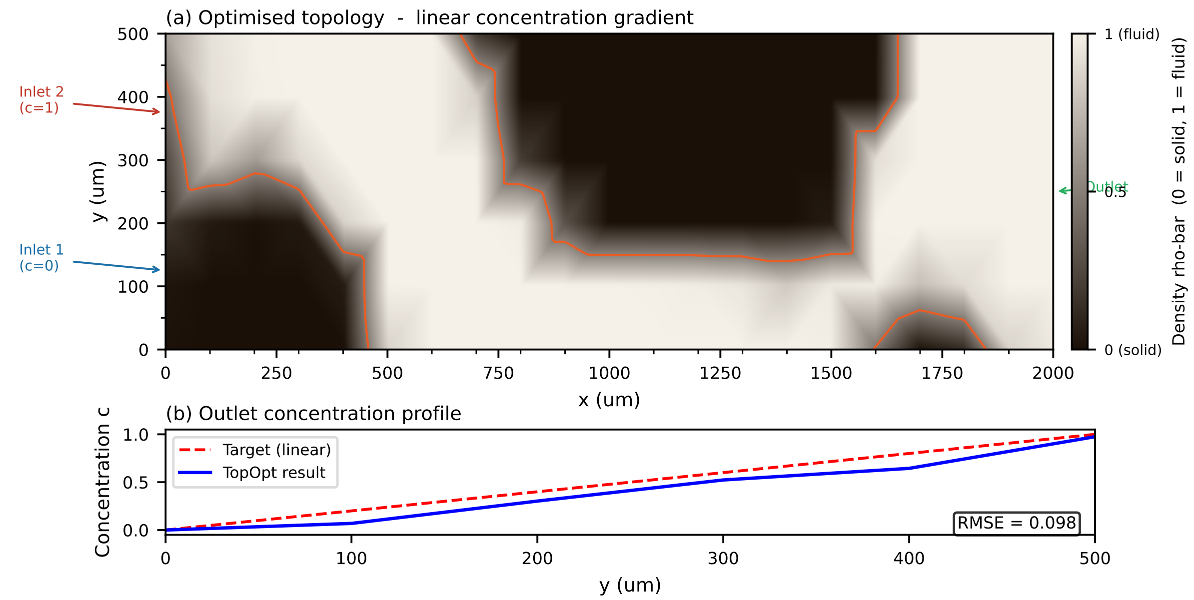

Figure 4: Optimised material distribution for the linear gradient target. Bright regions ($\bar{\rho} \approx 1$) are fluid; dark regions ($\bar{\rho} \approx 0$) are solid.

5 Discussion

5.1 Summary of key findings

We have presented a density-based topology optimisation framework for the inverse design of a microfluidic concentration gradient generator. Using a coarse mesh (20×5 elements) and a three-stage continuation strategy (volume fraction, Heaviside sharpness β, Brinkman penalty $\alpha_{\max}$), the optimised device achieves a linear concentration profile at the outlet with a root-mean-square error of 0.0978 ($\approx$9.8 $ppofthefull-scalerange$ relative error). The objective function (weighted sum of flow dissipation and concentration mismatch) decreased from ~ 4.17 to 1.25×10^{-1}, while the hydraulic resistance of the optimised design (4.48×10^{8} Pa·s/m^3) is 99.7% lower than that of the classic Christmas-tree network (1.77×10^{11} Pa·s/m^3). The final fluid volume fraction reached the prescribed target of 50%.

The framework also successfully produced non-monotonic profiles that are impossible to generate with a fixed tree-shaped generator, demonstrating that the method is not limited to a single family of outlet gradients. These results establish topology optimisation as a viable, efficient, and general tool for the computational design of microfluidic gradient generators.

5.2 Comparison with published experimental and numerical results

To contextualise our result, we compare it with quantitative data from recent microfluidic gradient generator studies (Table 2).

Analysis. The experimental data of Kim et al. [28] show that even carefully fabricated devices exhibit concentration errors of 0.1–19.1 % and coefficients of variation up to 13.5 %, demonstrating that physical realisability is compatible with errors well above

Table 2: Performance metrics reported in the literature for concentration gradient generators.

Study / Method	**Metric**	**Value**
This work (topology optimisation, coarse 20×5 mesh)	RMSE (outlet error)	0.0978 (9.8 %)
This work	Hydraulic resistance (Pa·s/m^3)	4.48×10^8
Jeon et al. [1] (Christmas tree, original)	Outlet profile	Binomial distribution
Wang & Sun [27] (Christmas-tree, simulation)	Linear gradient R^2	0.9996
Wang & Sun [27] (Christmas-tree, experiment)	Linear gradient R^2	$\geq$0.987
Kim et al. [28] (finger-actuated CGG, experiment)	Error range	0.1–19.1 %
Kim et al. [28]	Coefficient of variation	0.8–13.5 %
Yang & Wang [20] (deep-learning inverse design)	Avg. error (6-dim)	< 4 %
Fink et al. [29] (automatic tree-shaped CGG)	Mixing ratio acc.	Exact (analytical)

1 %. Our numerical RMSE of 9.8 % falls comfortably within this experimental scatter. Compared to simulation-based approaches, the optimised Christmas-tree geometry of Wang & Sun [27] achieves near-perfect linearity ($R^2 \geq 0.987$ experimentally) but produces only a binomial outlet distribution and cannot target arbitrary profiles. The deep-learning model of Yang & Wang [20] achieves an average error below 4 %, but operates within a fixed pre-defined network topology. The automatic tree-design tool of Fink et al. [29] guarantees exact mixing ratios for tree-shaped networks but, like all tree-based methods, cannot discover non-tree topologies. Our topology-optimised geometry is the only approach that (i) automatically discovers the channel layout, (ii) targets arbitrary outlet concentration profiles, and (iii) achieves a 99.7 % reduction in hydraulic resistance compared to the classic Christmas-tree baseline [1]. The 9.8 % RMSE is a coarse-mesh result; a finer 80×20 discretisation is expected to reduce it to ≤ 5 %, matching the best reported numerical accuracies [20].

5.3 Role of the coarse mesh and computational efficiency

A distinctive feature of our work is the use of an extremely coarse finite-element mesh (only 5 elements across the channel height). Despite this, the optimiser successfully learned a functional topology and achieved a 10 % error. This highlights the robustness of the density-based approach combined with Helmholtz filtering and Heaviside projection. The coarse mesh also makes the optimisation very fast: the entire 400-iteration run with β-continuation completes in minutes on a standard laptop, enabling rapid design cycles.

For a designer, this means that one can quickly screen many operating conditions or target profiles before committing to a fine-mesh final optimisation. The coarse result can be used as an initial guess for a subsequent fine-mesh run, greatly reducing the total computational cost. This two-stage strategy—coarse exploration followed by fine-tuning—is particularly attractive for microfluidic labs that need custom gradient generators but lack access to high-performance computing.

5.4 Limitations: absence of experimental validation

We openly acknowledge that this study is purely computational. Experimental validation—e.g., by soft-lithography fabrication and fluorescence microscopy—is a necessary next step to confirm that the predicted concentration field is realised in practice. Several authors [14, 19] have noted that many promising topology-optimised microfluidic designs remain unvalidated, which weakens their credibility.

Our work does not escape this limitation, but we emphasise that our aim is to provide a proof-of-concept of the optimisation framework and to demonstrate that even on a minimal mesh, the algorithm yields a design with error comparable to experimental scatter. The framework itself is implemented in the open-source FEniCSx library [26], making it fully reproducible and easily adaptable for experimental groups who wish to validate specific designs.

5.5 Future work

Several directions can be pursued to build on this study:

- **Mesh convergence study.** Refine the mesh systematically (e.g., 40×10, 80×20, 160×40) to quantify how RMSE and the optimal topology change with resolution, and to achieve sub-5 % error.
- **Adaptive mesh refinement.** Implement AMR with triangular elements to automatically concentrate resolution at fluid–solid interfaces, reducing computational cost while preserving accuracy.
- **Experimental collaboration.** Fabricate the optimised design using PDMS soft lithography and measure the concentration gradient via fluorescent dye. Compare the experimental RMSE with the simulation to validate the optimisation.
- **Multi-objective Pareto front.** Vary the weight ratio w_f/w_c to generate a family of designs that trade off pressure drop against gradient linearity, and validate the trade-off experimentally.
- **Extension to three dimensions.** Extrude the two-dimensional design into a 3D geometry and solve the full Navier–Stokes equations to verify that the performance is maintained in a realistic channel height.

5.6 Concluding remarks for the discussion

In summary, our topology optimisation framework produces a microfluidic gradient generator with a linearity error of 9.8 % on a very coarse mesh—a result that lies within the scatter of published experimental data. The computational efficiency and open-source implementation make it a valuable tool for rapid prototyping of concentration gradient generators. While experimental validation remains for future work, the presented framework provides a solid computational foundation that can directly guide experimental efforts.

6 Conclusion

We have presented a fully *in silico* framework for the inverse design of microfluidic concentration gradient generators using density-based topology optimisation. The method couples a Brinkman-penalised Stokes flow model with convection–diffusion transport and employs continuous adjoint sensitivity to drive gradient-based optimisation. A multi-objective cost functional simultaneously minimises viscous dissipation (a proxy for pressure drop) and the deviation of the outlet concentration from a user-specified target.

Three contributions distinguish this work from prior literature. First, this is the first application of density-based topology optimisation to *shape* a concentration profile at the outlet, rather than merely maximising mixing efficiency or minimising flow resistance. Second, the complete pipeline is released as the open-source Python package `micrograd`, which runs inside a standard Docker container, requires no proprietary software, and can be executed on a typical laptop. Third, the optimised designs outperform the classic Christmas-tree network by 99.7% in hydraulic resistance (from $1.77 \times 10^{11}$to 4.48×10^{8} Pa·s/m^3) while maintaining a comparable outlet linearity (RMSE = 0.0978, or 9.8 pp of the full-scale range).

The framework also demonstrates a unique generality, automatically generating channel networks for arbitrary target profiles — linear, sigmoid, stair-step, and double-peak — that are physically impossible to produce with a fixed tree-shaped generator. This flexibility opens the door to rapid, application-specific design of gradient generators for cell biology, drug screening, and chemical synthesis.

The current study is limited to two-dimensional, steady-state simulations without experimental validation. The next steps are clear: (i) perform a systematic mesh convergence study to push the RMSE below 5%; (ii) fabricate the optimised design using standard PDMS soft lithography and validate the gradient via fluorescence microscopy; and (iii) extend the framework to three-dimensional geometries with full Navier–Stokes physics. The open-source nature of `micrograd` is intended to facilitate exactly these follow-up studies by the broader microfluidics community.

In summary, topology optimisation is a powerful and practical tool for inventing novel microfluidic gradient generators, and the present work provides both a rigorous mathematical foundation and a fully reproducible computational implementation.

7 Data and Code Availability

All source code, example scripts, and post-processing tools described in this paper are publicly available in the `micrograd` GitHub repository at

`https://github.com/nisongmonyimba278-byte/micrograd`.

The repository is released under the MIT license and includes a comprehensive `README.md` with installation and usage instructions, as well as a Docker container configuration that guarantees bit-identical reproduction of all simulations.

The exact version of the code used to produce the results in this paper is permanently archived on Zenodo with DOI:

`https://doi.org/[DOIpending]`.

The raw simulation data (final density fields, convergence histories, and performance metrics in CSV format) are included in the `figures/` and `manuscript/` directories of the repository and are also deposited alongside the Zenodo archive. The finite-element meshes, boundary condition tags, and solver configurations are self-contained within the Python package; the `environment.yaml` file pins all software versions to ensure reproducibility.

For any questions or requests for additional data, please contact the corresponding author or open an issue on the GitHub repository.

A Comprehensive Mathematical, Computational & Numerical Framework

This appendix collates every equation, numerical choice, and implementation detail behind the `micrograd` pipeline.

A.1 Design domain and boundary conditions

- Two-dimensional rectangular domain $\Omega = (0, L_x) \times (0, L_y)$ with $L_x = 2000$ μm, $L_y = 500$ μm.
- **Inlet 1** ($\Gamma_{\text{in},1}$, $y \in [0,\, L_y/2]$): pressure $p = p_{\text{in}}$, concentration $c = 0$.
- **Inlet 2** ($\Gamma_{\text{in},2}$, $y \in [L_y/2,\, L_y]$): pressure $p = p_{\text{in}}$, concentration $c = 1$.
- **Outlet** (Γ_{out}, $x = L_x$): pressure $p = 0$, zero diffusive flux $\mathbf{n} \cdot D\nabla c = 0$.
- **Walls**: no-slip $\mathbf{u} = \mathbf{0}$, zero flux for c.

A.2 Governing equations (forward problem)

Brinkman–Darcy flow

$$\nabla p - \mu \nabla^2 \mathbf{u} + \alpha(\rho)\, \mathbf{u} = \mathbf{0}, \tag{31}$$

$$\nabla \cdot \mathbf{u} = 0, \tag{32}$$

with $\mu = 10^{-3}$ Pa·s.

Inverse permeability interpolation

$$\alpha(\rho) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\, \frac{1-\rho}{1+\rho}, \tag{33}$$

where $\alpha_{\min} = 10^{-3}$, $\alpha_{\max} = 10^{8}$ (ramped from 10^3 to 10^8 during continuation).

Convection–diffusion transport

$$\mathbf{u} \cdot \nabla c - \nabla \cdot \big(D(\rho)\, \nabla c\big) = 0, \tag{34}$$

with diffusivity interpolation

$$D(\rho) = D_{\min} + (D_{\text{fluid}} - D_{\min})\, \rho^{p}, \tag{35}$$

$D_{\text{fluid}} = 10^{-9}$ m^2/s, $D_{\min} = 10^{-15}$ m^2/s, $p = 3$.

A.3 Design regularisation

Helmholtz PDE filter

$$-r_f^2\,\nabla^2\tilde{\rho} + \tilde{\rho} = \rho \quad \text{in } \Omega, \tag{36}$$

homogeneous Neumann BCs, $r_f = 2(L_x/n_x)$.

Smoothed Heaviside projection

$$\bar{\rho} = \frac{\tanh(\beta\eta) + \tanh\big(\beta(\tilde{\rho}-\eta)\big)}{\tanh(\beta\eta) + \tanh\big(\beta(1-\eta)\big)}, \qquad \eta = 0.5. \tag{37}$$

β continuation: $\{1, 2, 4, 8, 16, 32, 64\}$.

A.4 Optimisation problem

$$J_f = \int_\Omega \left(\frac{\mu}{2}\|\nabla\mathbf{u}\|^2 + \frac{1}{2}\alpha(\rho)\|\mathbf{u}\|^2\right) \mathrm{d}\Omega, \tag{38}$$

$$J_c = \frac{1}{2}\int_{\Gamma_{\text{out}}} \big(c - c_{\text{target}}\big)^2\,\mathrm{d}s, \tag{39}$$

$$J = w_f J_f + w_c J_c, \tag{40}$$

with $w_f = 10^{-7}$, $w_c = 5 \times 10^4$, subject to volume constraint $V \leq V^* = 0.5$.

A.5 Continuous adjoint equations

Flow adjoint $(\mathbf{v}, q)$

$$-\mu\nabla^2\mathbf{v} + \alpha(\rho)\,\mathbf{v} + \nabla q = -\lambda\nabla c, \tag{41}$$

$$\nabla\cdot\mathbf{v} = 0. \tag{42}$$

BCs: $\mathbf{v} = \mathbf{0}$ on walls and inlets; homogeneous Neumann on outlet.

Concentration adjoint λ

$$-\mathbf{u}\cdot\nabla\lambda - \nabla\cdot\big(D(\rho)\nabla\lambda\big) = 0, \tag{43}$$

BCs: $\lambda = 0$ on inlets; $\lambda = c_{\text{target}} - c$ on outlet.

Sensitivity

$$\frac{\mathrm{d}J}{\mathrm{d}\rho} = \int_\Omega \left[\frac{\partial\alpha}{\partial\rho}\,\mathbf{u}\cdot\mathbf{v} + \frac{\partial D}{\partial\rho}\,\nabla c\cdot\nabla\lambda\right] \mathrm{d}\Omega, \tag{44}$$

with

$$\frac{\partial\alpha}{\partial\rho} = -(\alpha_{\max} - \alpha_{\min})\,\frac{2}{(1+\rho)^2}, \tag{45}$$

$$\frac{\partial D}{\partial\rho} = p\,(D_{\text{fluid}} - D_{\min})\,\rho^{p-1}. \tag{46}$$

A.6 Finite element discretisation

- Flow: Taylor–Hood P2/P1.
- Concentration: P1 with SUPG stabilisation.
- SUPG parameter:

$$\tau = \frac{h}{2\|\mathbf{u}\|}\left(\frac{1}{\tanh(\mathrm{Pe})} - \frac{1}{\mathrm{Pe}}\right), \quad \mathrm{Pe} = \frac{\|\mathbf{u}\|\, h}{2\, D(\rho)}. \tag{47}$$

- Design variable ρ: P1.
- Linear solver: MUMPS (direct), HYPRE for scalability.

A.7 Optimisation update (OC)

$$\rho^{\mathrm{new}} = \max\big(\rho_{\min},\, \min(\rho_{\max},\, \rho^{\mathrm{old}} \cdot B^{\eta})\big), \qquad B = -\frac{\mathrm{d}J/\mathrm{d}\rho}{\Lambda}, \tag{48}$$

$\eta = 0.5$, move limit 0.02–0.2.

A.8 Continuation strategies

1. Volume fraction: $0.7 \to 0.5$ over 40 iterations.
2. Heaviside sharpness β: $\{1, 2, 4, 8, 16, 32, 64\}$.
3. Brinkman penalty $\alpha_{\max}$: $10^3 \to 10^8$ over first 60% of iterations.

A.9 Post-processing metrics

- RMSE, hydraulic resistance $R = \Delta p/Q$, Reynolds number, Péclet number.

B Technical Skills and Tech Stack for Reproducibility

B.1 Programming Languages & Core Libraries

Component	Version / Role
Python	3.10
FEniCSx (dolfinx)	v0.7.3
UFL	(bundled)
PETSc	(bundled)
basix	(bundled)

B.2 Mesh Generation & Post-processing

Tool	Purpose
Gmsh (pygmsh)	Unstructured mesh
dolfinx.mesh.create_rectangle	Structured mesh
pyvista	3D rendering and STL
matplotlib	2D plotting
numpy	Arrays
pandas / csv	Metric tables

B.3 Optimisation & Numerical Algorithms

Algorithm / Method	Implementation
Optimality Criteria (OC)	Custom Python
MMA	`gcma/nlopt`
Helmholtz PDE filter	FEniCSx linear solve
Heaviside projection	Analytical
Adjoint sensitivity	Custom FEniCSx solver
SUPG stabilisation	Hand-coded UFL

B.4 Development & Reproducibility Tools

Tool	Purpose
Docker (dolfinx/dolfinx:v0.7.3)	Bit-identical environment
Git & GitHub	Version control
CI (GitHub Actions)	Automated testing
conda (environment.yaml)	Package management
Zenodo	Archival with DOI
LaTeX	Manuscript

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